Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/28/2006
PI Name:	Munir, Zuhair A Ph.D.		
Project Title:	Electric Field Effects in Self-Propagating High-Temperature Combustion Synthesis under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Shaw, Benjamin, D. Ph.D. (Department of Mechanical & Aeronautical Engineering, University of California, Davis)		
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	The research will investigate the effect of an electric field on the dynamics of self-propagating high-temperature synthesis (SHS) waves and the effect of the field on phase formation and structure evolution in SHS. The planned work includes both terrestrial and reduced gravity experiments. The latter will be carried out in parabolic flights and in drop towers. The parabolic flight experiments will use existing facilities developed by Guigne International, Ltd. in NASA's KC-135 aircraft. Modifications were done to accommodate the requirements for application of an electric field application as well as instrumentation for temperature measurements and imaging. Drop tower experiments will be carried out in the NASA Glenn Research Center 2.2 s Drop Tower, and additional drop tower experiments will be carried out in Japan by one of the collaborating scientists. The research will employ systems that produce varied amounts of liquid. For example, the systems Si + C or Ti + Al allow investigation of the case where one reactant melts and the other does not, the system Ta + C allows investigation of a system where neither the reactants not the products melt, and the system FeAl allows investigation of the case where both reactants and products melt. These systems are expected to		

Task Description:	respond to the gravity level in different ways, e.g., through liquid transport effects. It is also planned that quenching experiments will be performed. This will be accomplished by turning off the electric field during the combustion process. The removal of the field will cause the deflagration wave to quench under certain conditions, essentially freezing the structure of the wave. Wave structure will be analyzed using XRD, SEM and EPMA. We plan to quantify heat losses from samples using several techniques. We will also pursue computational and asymptotic modeling efforts. The computational efforts involve extending previous numerical models to account for phenomena such as melting, capillary flows, buoyant flows and three spatial dimensions. Asymptotic studies will be concerned with determining influences of electric fields on stability and extinction of field-assisted SHS waves, taking into account capillary transport of liquids as well as local variations in surface tension and contact angles from thermal or electrical effects. This research program is focused on studying self-propagating high-temperature synthesis (SHS) under reduced gravity conditions. The broad objective of this research program is to understand the role of an electric field in SHS reactions under conditions where gravity-related effects are suppressed. The research will allow increased understanding of fundamental aspects of field-assisted SHS processes as well as synthesis of materials that cannot be formed in normal gravity. For example, the electric field has been shown to influence the nature of the product, the composition of phases in a composite, and the homogeneity (elemental distribution) in solid solutions. Furthermore, the field can influence other important SHS-related phenomena including capillary flow, mass-transport in porous media, and Marangoni flows. These phenomena are influenced by gravity in conventional SHS processes (i.e., without electric fields). As a result the influence of the field on SHS under reduced gravity is expected to be different than under normal gravity. It is also known that heat loss rates from samples, which can depend significantly on gravity, can influence final products in SHS. This research will provide new information on the effects of gravity on self-propagating high-temperature combustion synthesis with applied electric fields. The research will increase understanding of the effects of buoyancy, segregation, percolation and may potentially lead to new materials that cannot be formed under normal gravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The reactive settling in the system Al-W was investigated under different gravitational conditions and under the influence of an electric field. Compacted powder mixtures of 0.1W + 1.0 Al were reacted inside chemical ovens under conditions of 0 g (low gravity), 0.16 g (lunar gravity), 1.0 g (terrestrial gravity), and 1.8 g (high gravity). The chemical ovens selected were Nb + 2Si and Ti + Al which provided different temperatures. The elements reacted partially to form WAl4 under 0, 0.16, and 1.8 g conditions aboard NASA KC-135 plane. The effect of gravity on the microstructural distribution of the phases WAl4 and unreacted W and Al was investigated. In the 0 g case the unreacted W was randomly distributed throughout the sample and under 0.16 g partial settling of the W particles was observed. When the reaction was carried out at 1 g, the settling of W particles is virtually total. Similar microstructural distribution was observed in the case of 1.8 g. Settling of the WAl4 intermetallic phase was not observed even at the highest gravity condition. Moreover, the applied electric current did not affect the settling or the distribution of W grains; similar microstructures were found for each gravity condition with and without the electric current. The settling pattern showed no dependence on the direction of the combustion wave; bottom and top ignition samples exhibited similar microstructures. The reactive settling process was carried out under the influence of an electric field. It was seen that the field had no discernable effect on the resulting microstructure. Furthermore in the Al-W system, the dissolution and precipitation was investigated. Once the reaction temperatures are reached, W will dissolve into liquid Al. In turn, WAl4 precipitates were found to form upon cooling of the sample. The precipitates grew larger with increasing gravity, due to the absence of unreacted W particles in 0.16, 1, and 1.8 g samples. Therefore gravity has been shown to affect the precipitate size indirectly. Electric current showed little effect on the precipitate length in terrestrial gravity. On the other hand, in microgravity, samples reacted at 200A and 350A showed about 40% and 36% increases in precipitate size, respectively. This is believed to be due to the increased availability of dissolved W in the 200A and 350A samples and due to the possibility of enhanced mass transport by electric current. Neither electric current nor gravity was found to affect the alignment, directionality, and distribution of the intermetallic phase. It was found that increasing gravity resulted in an increase in the dissolution of W, resulting in the formation of higher amounts of WAl4. This is believed to be due to an increase in dissolution rates created by the stirring effect of the settling motion of the unreacted W particles. Finally, increasing electric current was observed to enhance the W dissolution rate into liquid Al. Although the chemical oven combustion waves moved towards the top (warmer and less dense liquid at the bottom), unstable convection is shown to be absent even at the highest gravity condition. In addition to the Al-W system, the possibility of settling was investigated in terms of density differences between the solid and the liquid phase, solubility and dissolution rate (temperature), and initial solid concentration for Ta, Ti and Ni in liquid Al. Although settling in isothermal systems is a well-understood phenomenon, it is quite difficult per above-mentioned parameters in the dynamic conditions present in SHS reactions. As previously mentioned in the W-Al system settling was only observed at high reaction temperatures when the Nb + 2Si chemical oven was used. At low temperatures the formation of the intermetallic phase (WAl4) was observed to prevent settling. On the other hand, for Ta and Ti settling was only observed at low reaction temperatures when the Ti + Al chemical oven was used. At high temperatures total dissolution was observed for Ta and Ti. Settling of Ti was found to depend on the grain size (Ti) due to the lower density differences between Ti and liquid Al. Total dissolution was the case for Ni even at the lowest temperature experimented. Only the intermetallic precipitates in the Ta-Al system showed settling behavior. The reactive settling behavior can be explained with the corresponding solubilities at different temperatures. Reactive capillary infiltration experiments were performed in porous samples inside of thin graphite foil capsules, using the chemical oven technique. The capsules consisted of a single layer of Al and a single layer of Ni or Ti. In order to investigate the gravitational effects of the movement of the liquid, the Al was placed on the bottom or the top of the Ni or Ti. Experiments were performed in 0 and 1 g with and without electric current (0 and 350 A). Capillary infiltration distances were measured for the experiments under 0 A and were found to be larger under 1 g conditions for all four experimental cases. However, the distance of infiltration of the Al into the Ti was greater than into the Ni. This is contradictory to equilibrium contact angles reported in the literature. On the other hand, SHS reactions are, for the most part, under non-equilibrium conditions therefore such values should be considered. Experiments under 350 A showed complete infiltration of the liquid Al into the Ni or Ti for both 0 and 1 g.

Bibliography Type:	Description: (Last Updated: 02/27/2006)
Articles in Peer-reviewed Journals	Unuvar, C.; Fredrick, D.M.; Anselmi-Tamburini, U. ; Manerbino, A.; Guigne, J.Y.; Shaw, B.D.; Munir, Z.A. "Reactive Settling in the Al-W System" submitted to Intermetallics , Mar-2006
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